# CENG381 Stochastic Processes

Biased Random Walk — Practice 1 (Answer Key)

## Q1 Solution

### a) Recurrence and boundary conditions:

```
Rn = q*R(n-1) + p*R(n+1)   for 1 <= n <= 3
   = (1/3)*R(n-1) + (2/3)*R(n+1)

R0 = 0  (Pit of Disaster is the losing state)
R4 = 1  (Home is the winning state)
```

### b) Characteristic equation and roots:

```
p*r^2 - r + q = 0
(2/3)*r^2 - r + (1/3) = 0  =>  multiply by 3:
2r^2 - 3r + 1 = 0  =>  (2r-1)(r-1) = 0
r1 = 1/2 = q/p,   r2 = 1

General solution:  Rn = alpha1*(1/2)^n + alpha2
```

### c) Applying boundary conditions and computing $R_2$:

```
R0 = 0:  alpha1 + alpha2 = 0  =>  alpha2 = -alpha1
R4 = 1:  alpha1*(1/16) - alpha1 = 1
         alpha1*(1/16 - 1) = 1
         alpha1*(-15/16)  = 1
         alpha1 = -16/15,   alpha2 = 16/15

Rn = (16/15)*[1 - (1/2)^n]

R2 = (16/15)*(1 - 1/4) = (16/15)*(3/4) = 48/60 = 4/5 = 0.80

Unbiased:  R2 = 1/2 = 0.50
```

The rightward drift (p=2/3) raises the win probability from 50% to 80%. Since p > q, the walk tends to drift towards the winning boundary, so $R_2$ increases above the unbiased value.

## Q2 Solution

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## a) Recurrence for $X_n$:

```
Xn = 1 + q*X(n-1) + p*X(n+1)
   = 1 + (1/3)*X(n-1) + (2/3)*X(n+1)   for 1 <= n <= 3
X0 = 0,   X4 = 0
```

## b) Particular solution cn + d:

Substitute $X_n = cn + d$ into the recurrence:

```
cn+d = 1 + (1/3)*[c(n-1)+d] + (2/3)*[c(n+1)+d]
cn+d = 1 + cn/3 - c/3 + d/3 + 2cn/3 + 2c/3 + 2d/3
cn+d = 1 + cn*(1/3+2/3) + c*(-1/3+2/3) + d*(1/3+2/3)
cn+d = 1 + cn + c/3 + d

Coefficient of n:  0 = 1 + c/3  =>  c = -3
Constant:          0 = 1  =>  contradiction if d is free;
                   set d = 0 (absorbed into alpha2).

Particular solution:  Xn^(p) = -3n
(Equivalently: c = -1/(p-q) = -1/(2/3-1/3) = -1/(1/3) = -3  ✓)
```

## c) Full solution and $X_2$:

```
Xn = alpha1*(1/2)^n + alpha2 - 3n

X0=0:  alpha1 + alpha2 = 0  =>  alpha2 = -alpha1
X4=0:  alpha1*(1/16) - alpha1 - 12 = 0
       alpha1*(1/16 - 1) = 12
       alpha1*(-15/16) = 12
       alpha1 = -12*(16/15) = -64/5
       alpha2 = 64/5

Xn = (64/5)*[1 - (1/2)^n] - 3n

X2 = (64/5)*(1 - 1/4) - 6
   = (64/5)*(3/4) - 6
   = 48/5 - 30/5 = 18/5 = 3.6 steps
```

Compare with the unbiased expected time $X_2 = 2 \cdot (4-2) = 4$ steps. The biased walk reaches absorption faster (3.6 < 4) because the rightward drift moves the particle efficiently toward state 4.

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## Q3 Solution

### a) Probability of reaching Cliff of Doom (state 6), $R_0=0$, $R_6=1$:

```
q/p = (1/3)/(2/3) = 1/2,    w = 6

Rn = [(1/2)^n − 1] / [(1/2)^6 − 1]
   = [(1/2)^n − 1] / (1/64 − 1)
   = [(1/2)^n − 1] / (−63/64)
   = (64/63)*[1 − (1/2)^n]

R3 = (64/63)*(1 − 1/8) = (64/63)*(7/8) = 448/504 = 8/9
```

$R_3 = 8/9 \approx 89\%$. The particle is much more likely to reach the Cliff of Doom (state 6) than the Pit of Disaster (state 0), because p=2/3 creates a strong rightward bias.

### b) Probability of reaching Pit of Disaster (state 0), $R_0=1$, $R_6=0$:

```
Rn = [(1/2)^n − (1/2)^6] / [1 − (1/2)^6]
   = [(1/2)^n − 1/64] / [1 − 1/64]
   = [(1/2)^n − 1/64] / (63/64)

R3 = [(1/8) − (1/64)] / (63/64)
   = [(8/64) − (1/64)] * (64/63)
   = (7/64) * (64/63) = 7/63 = 1/9

Check:  R3(Cliff) + R3(Pit) = 8/9 + 1/9 = 1  ✓
```

### c) Expected absorption time $X_3$ and comparison:

```
Xn = alpha1*(1/2)^n + alpha2 − 3n  [same particular solution]

X0=0:  alpha1 + alpha2 = 0  =>  alpha2 = −alpha1
X6=0:  alpha1*(1/64) − alpha1 − 18 = 0
       alpha1*(1/64 − 1) = 18
```

```
       alpha1*(-63/64) = 18
       alpha1 = -18*(64/63) = -1152/63 = -128/7
       alpha2 = 128/7

Xn = (128/7)*[1 - (1/2)^n] - 3n

X3 = (128/7)*(1 - 1/8) - 9
   = (128/7)*(7/8) - 9
   = 128/8 - 9 = 16 - 9 = 7 steps
```

Biased: $X_3$ = 7 steps. Unbiased: $X_3 = n(w-n) = 3 \cdot 3 = 9$ steps. The biased walk ($p=2/3$) is absorbed 2 steps faster on average. This makes sense: the drift toward the right boundary shortens the time to reach either absorbing state.